L05: Support Vector Machines

CSci 574 Machine Learning (Géron) Ch. 5

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Fall 2025

Large Margin Classification

- When two classes can be separated by a line (hyperplane). Figure 1, left
  - They are **linearly separable**
- SVM classifier fits the widest possible street. Figure 1, right
  - This is called **large margin classification**

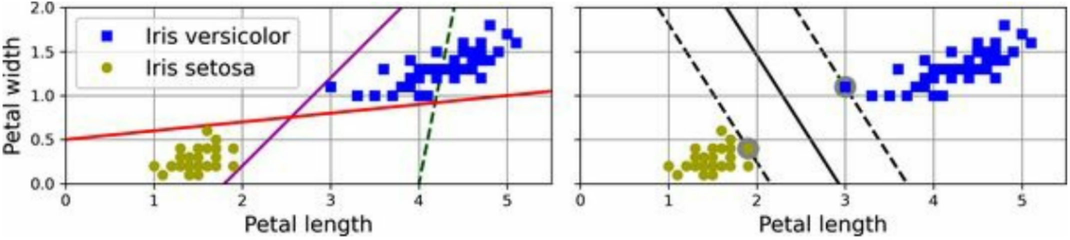

Figure 1: Large margin classification (Géron, 2023, pg.287)

Linear Support Vector Machine

- Notice that adding more training instances “off the street” will not affect the decision boundary at all.
  - It is fully determined (or “supported”) by the instances located on the edge of the street
  - These instances are called the **support vectors**
- A Support Vector Machine (SVM) is a machine learning model capable of performing linear or nonlinear classification and regression.
- It works by identifying support vectors to determine the largest margin decision boundary

SVMs are Sensitive to Scaling

- In the left plot (Figure 2) vertical scale is much larger than the horizontal scale, gives a almost horizontal large margin with these support vectors.
- In the right plot (Figure 2) after standard feature scaling, the decision boundary had a much different slope.

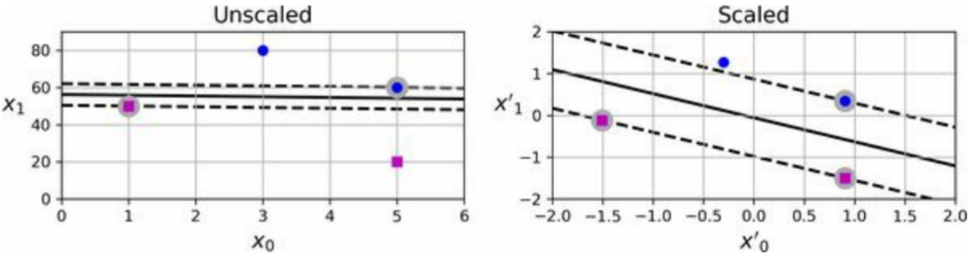

Figure 2: SVM sensitivity to feature scales. (Géron, 2023, pg.288)

Soft Margin Classification

- If we strictly require all instances off street and on correct side of margin, this is called **hard margin classification**.
- Only possibly when data are linearly separably.
  - Figure 3 left, impossible to find hard margin, not linearly separable.
- Also sensitive to outliers.
  - Figure 3 right, decision boundary ends up very different if there is an outlier here, might not generalize.

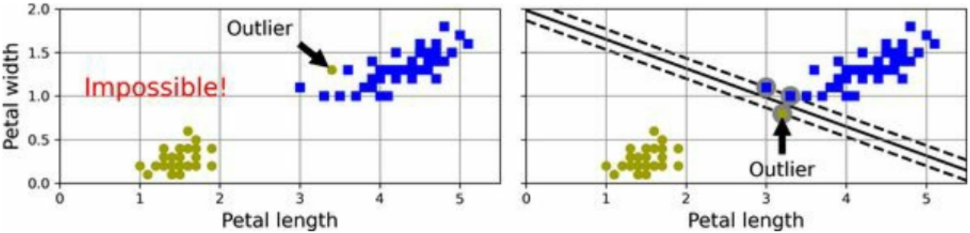

Figure 3: Hard margin sensitivity to outliers. (Géron, 2023, pg.288)

Soft Margin Classification

- To avoid both, find balance between keeping street as large as possible and limiting the **margin violations**
  - This is **soft margin classification**
- SVM use regularization hyperparameter `C`
  - When `C` is low, large margin, more margin violations allowed, but if too low can cause underfitting.
  - When `C` is high, small margins, less margin violations allowed, but can overfit and be sensitive to outliers

SVM C Regularization

If your SVM model is overfitting, you can regularize it by reducing C, leading to softer margins which will tend to be better at generalizing performance to unseen data.

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Bibliography

Géron, A. (2023). *Hands-on machine learning with scikit-learn, keras and tensorflow* (third). O'Reilly Media, Inc.